

# Notation

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Mathematical notation is not standardized. This section states the conventions these notes use, and the alternatives you may meet in other sources.

Table 1: Notation used in this book

| symbol                           | meaning                                                                  | LaTeX                                       |
|----------------------------------|--------------------------------------------------------------------------|---------------------------------------------|
| $\neg$                           | not                                                                      | <code>\neg</code>                           |
| $\forall$                        | all                                                                      | <code>\forall</code>                        |
| $\exists$                        | some                                                                     | <code>\exists</code>                        |
| $\cup$                           | union, “or”                                                              | <code>\cup</code>                           |
| $\cap$                           | intersection, “and”                                                      | <code>\cap</code>                           |
| $ $                              | given, conditional on                                                    | <code>\mid,  </code>                        |
| $\sum$                           | <a href="#">sum</a>                                                      | <code>\sum</code>                           |
| $\prod$                          | product                                                                  | <code>\prod</code>                          |
| $\mu$                            | mean                                                                     | <code>\mu</code>                            |
| $\mathbb{E}$                     | <a href="#">expectation</a>                                              | <code>\mathbb{E}</code>                     |
| $x^\top$                         | transpose of $x$                                                         | <code>x^{\top}</code>                       |
| $'$                              | transpose or <a href="#">derivative</a> <sup>1</sup>                     | <code>'</code>                              |
| $\perp$                          | <a href="#">independent</a>                                              | <code>\perp</code>                          |
| $\therefore$                     | therefore, thus                                                          | <code>\therefore</code>                     |
| $\eta$                           | <a href="#">linear component of a GLM</a>                                | <code>\eta</code>                           |
| $\lfloor x \rfloor$              | floor of $x$ : largest <a href="#">integer</a> less than or equal to $x$ | <code>\lfloor x \rfloor</code>              |
| $\lceil x \rceil$                | ceiling of $x$ : smallest integer greater than or equal to $x$           | <code>\lceil x \rceil</code>                |
| $\mathbb{1}_A(x), \mathbb{1}(P)$ | indicator function (Section ): 1 if condition holds, 0 otherwise         | <code>\mathbb{1}_A(x), \mathbb{1}(P)</code> |

<sup>1</sup>depending on whether it is applied to a matrix or a function

## Writing math in Quarto

The third column of Table 1 gives the LaTeX command for each symbol. Quarto and R Markdown documents write math in LaTeX syntax: put inline math between single dollar signs, as in  $x^2$ , and displayed equations between double dollar signs, as in 
$$x^2$$
. The [equations section of Quarto's Markdown guide](#) shows the details. These notes also use shorthand macros, such as `\floor{x}` for  $\lfloor x \rfloor$ , defined in the `latex-macros` submodule's `macros.qmd`.

## Natural numbers

**Exercise 0.1** (Which numbers are natural?). List the elements of the set  $\{n \in \mathbb{N} : n < 3\}$ . Is your answer the same in every textbook?

Solution

*Solution 0.1.* The answer depends on whether the source counts 0 as a natural number:

- if  $\mathbb{N}$  starts at 0, the set is  $\{0, 1, 2\}$ ;
- if  $\mathbb{N}$  starts at 1, the set is  $\{1, 2\}$ .

Both conventions are in common use, so the answer is not the same in every textbook.

**Definition 0.1** (Integers). The **integers** are the whole numbers, together with their negatives and 0:

$$\mathbb{Z} \stackrel{\text{def}}{=} \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$$

An **integer** is an element of  $\mathbb{Z}$ . The integers greater than 0 are the **positive integers**, and the integers less than 0 are the **negative integers**.

**Example 0.1** (Integers and non-integers).

- $-7$ ,  $0$ , and  $12$  are integers.
- $\frac{1}{2}$  and  $-2.5$  are not integers.
- $\frac{6}{3}$  is an integer, because  $\frac{6}{3} = 2$ .

**Definition 0.2** (Even and odd integers). An integer (Definition 0.1)  $n$  is **even** if  $n = 2k$  for some integer  $k$ , and **odd** if  $n = 2k + 1$  for some integer  $k$ .

**Example 0.2** (Even and odd integers).

- 6 is even, because  $6 = 2 \cdot 3$ .
- 0 is even, because  $0 = 2 \cdot 0$ .
- $-3$  is odd, because  $-3 = 2 \cdot (-2) + 1$ .
- 7 is odd, because  $7 = 2 \cdot 3 + 1$ .
- 2.5 is neither even nor odd, because it is not an integer.

**Definition 0.3** (Natural numbers (our convention)). In these notes, the **natural numbers** are the positive integers (Definition 0.1):

$$\mathbb{N} \stackrel{\text{def}}{=} \{1, 2, 3, \dots\}$$

**Definition 0.4** (Non-negative integers). The **non-negative integers** are the natural numbers (Definition 0.3) together with 0:

$$\mathbb{N}_0 \stackrel{\text{def}}{=} \{0, 1, 2, 3, \dots\} = \mathbb{N} \cup \{0\}$$

**Example 0.3** (Natural numbers in a data analysis).

- Observation indices start at 1, so we write  $i \in \{1, \dots, n\}$  with  $n \in \mathbb{N}$ .
- A count outcome, such as the number of hospital visits in a year, can be 0, so its support is  $\mathbb{N}_0$ , not  $\mathbb{N}$ .

### **i** Other conventions for the natural numbers

Other sources may define  $\mathbb{N}$  differently, so check each source's definition before reading its formulas:

- Many sources include 0 in  $\mathbb{N}$ . The international standard ISO 80000-2 defines  $\mathbb{N}$  to include 0, continuing the earlier standard ISO 31-11 (1978).
- Other sources start  $\mathbb{N}$  at 1, as these notes do.
- To remove the ambiguity, some sources write  $\mathbb{N}_1$  or  $\mathbb{Z}^+$  for  $\{1, 2, 3, \dots\}$ , and  $\mathbb{N}_0$  or  $\mathbb{Z}^{0+}$  for  $\{0, 1, 2, \dots\}$ .
- The words vary too. “Positive integers” ( $\{1, 2, 3, \dots\}$ ) and “non-negative integers” ( $\{0, 1, 2, \dots\}$ ) are unambiguous. “Whole numbers” usually includes 0, but can also mean all of the integers, negative ones included. “Counting numbers” usually starts at 1, but some sources include 0.
- Some older texts write  $J$  for the natural numbers.

When a formula's meaning depends on whether 0 is included, we write the set out explicitly, for example  $\{0, 1, 2, \dots\}$  or  $\{1, \dots, n\}$ .

Source: [Wikipedia](#), “[Natural number](#)”, “[Terminology and notation](#)” and “[Zero as natural number](#)”, which cites ISO 80000-2:2019 and the textbooks using each notation.

## Rational, real, and irrational numbers

**Definition 0.5** (Rational numbers). A **rational number** is a number that can be written as a fraction  $\frac{a}{b}$ , where  $a$  and  $b$  are integers (Definition 0.1) and  $b \neq 0$ . The set of rational numbers is written  $\mathbb{Q}$ :

$$\mathbb{Q} \stackrel{\text{def}}{=} \left\{ \frac{a}{b} : a \in \mathbb{Z}, b \in \mathbb{Z}, b \neq 0 \right\}$$

**Example 0.4** (Rational numbers).

- $\frac{3}{4}$  is rational, with  $a = 3$  and  $b = 4$ .
- 0.75 is rational, because  $0.75 = \frac{3}{4}$ .
- Every integer is rational: for example,  $-5 = \frac{-5}{1}$ .
- 0.333..., with the 3s repeating forever, is rational, because  $0.333\dots = \frac{1}{3}$ .

**Definition 0.6** (Real numbers). A **real number** is a number that can be written as a decimal expansion: an integer part, followed by a decimal point and a finite or infinite list of decimal places after it. Equivalently, the real numbers are the points on the number line. The set of real numbers is written  $\mathbb{R}$ .

*Remark 0.1* (A working definition). Definition 0.6 is a working definition, not a construction: it does not say what an infinite decimal expansion is, or how to add or multiply two of them. Rudin (1976, Appendix to Chapter 1) constructs the real numbers rigorously. Some real numbers have two decimal expansions: for example, 0.999... and 1.000... are the same real number, 1.

**Example 0.5** (Real numbers).

- 3,  $-0.5$ , and  $\frac{1}{3} = 0.333\dots$  are real numbers.
- $\pi = 3.14159\dots$  is a real number.
- Every rational number (Definition 0.5) is a real number, so  $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$ .

**Definition 0.7** (Irrational numbers). An **irrational number** is a real number (Definition 0.6) that is not rational (Definition 0.5).

**Example 0.6** (Irrational numbers).

- $\sqrt{2} = 1.41421 \dots$ , the [square root](#) of 2, is irrational: no fraction of integers  $\frac{a}{b}$  squares to 2 (see [Wikipedia: Square root of 2](#)).
- $\pi = 3.14159 \dots$  is irrational (see [Wikipedia: Proof that  \$\pi\$  is irrational](#)).
- 1.5 is not irrational, because  $1.5 = \frac{3}{2}$  is rational.

## Percent sign (“%”)

The percent sign “%” is just a shorthand for “/100”. The word “percent” comes from the Latin “per centum”; “centum” is Latin for 100, so “percent” means “per hundred” (cf. <https://en.wikipedia.org/wiki/Percentage>)

So, contrary to what you may have learned previously,  $10\% = 0.1$  is a true and correct equality, just as  $10\text{kg} = 10,000\text{g}$  is true and correct.

**i** Proof

*Proof.*

$$\begin{aligned} 10\% &= 10/100 \\ &= \frac{10}{100} \\ &= 0.1 \end{aligned}$$

□

You are welcome to switch between decimal and percent notation freely; just make sure you execute it correctly.

## Proofs

We can use any of:

- $\therefore$  (`\therefore` in LaTeX),
- $\Rightarrow$  (`\Rightarrow`),
- $\models$  (`\models`)

to denote logical entailments (deductive consequences).

Let's save  $\rightarrow$  (`\rightarrow`) for [convergence](#) results.

**Definition 0.8** (Proposition). A **proposition**, or **statement**, is a sentence that is either true or false.

**Example 0.7** (Propositions, and a sentence that is not one).

- “ $2 + 3 = 5$ ” is a proposition, and it is true.
- “7 is [even](#)” is a proposition, and it is false.
- “ $x > 2$ ” is not a proposition on its own: it is true when  $x = 3$  and false when  $x = 1$ , so it has no truth value until  $x$  is given a value.

**Definition 0.9** (Predicate). A **predicate**  $P(x)$  is a sentence about a variable  $x$  that becomes a proposition (Definition 0.8) whenever a value is substituted for  $x$ .

**Example 0.8** (A predicate). Let  $P(x)$  be the predicate “ $x > 2$ ”, for real numbers  $x$ .

- $P(3)$  is the proposition “ $3 > 2$ ”, which is true.
- $P(1)$  is the proposition “ $1 > 2$ ”, which is false.

**Definition 0.10** (Proof). A **proof** of a proposition (Definition 0.8) is a finite list of statements that ends with that proposition, in which each statement is

- an assumption,
- a definition,
- a previously proved result, or
- a consequence of earlier statements in the list, by one rule of logic or algebra.

**Example 0.9** (The sum of two even integers is even). Let  $a$  and  $b$  be even integers (Definition 0.2). Then  $a = 2j$  and  $b = 2k$  for some integers  $j$  and  $k$ , and

$$\begin{aligned} a + b &= 2j + 2k && \text{(substitute } a = 2j \text{ and } b = 2k) \\ &= 2(j + k) && \text{(distributive law)} \end{aligned}$$

The second line uses the [distributive law](#). Since  $j + k$  is an integer,  $a + b$  is even (Definition 0.2). For example,  $4 + 10 = 2 \cdot 2 + 2 \cdot 5 = 2 \cdot 7 = 14$ .

**Definition 0.11** (Result, theorem, lemma, and corollary). A **result** is a proposition (Definition 0.8) that has a proof (Definition 0.10). These notes label each result by its role:

- a **theorem** is a result of interest in its own right;
- a **lemma** is a result proved mainly as a step in the proof of another result;
- a **corollary** is a result that follows from a theorem with little extra work.

*Remark 0.2* (“Proposition” as a kind of result). Some sources also call a result of moderate importance a “proposition”. In these notes, “proposition” always means a sentence that is true or false (Definition 0.8), whether or not it has been proved.

**Example 0.10** (A corollary of a result). Example 0.9 proves the result “the sum of two even integers is even”. A corollary follows with little extra work: the sum of three even integers  $a$ ,  $b$ , and  $c$  is even, because  $a + b + c = (a + b) + c$ ,  $a + b$  is even by that result, and applying the result again to the even integers  $a + b$  and  $c$  shows that  $(a + b) + c$  is even. For example,  $2 + 4 + 6 = (2 + 4) + 6 = 6 + 6 = 12$ , which is even.

**Definition 0.12** (Counterexample). A **counterexample** to a proposition of the form “for every  $x$  in  $A$ ,  $P(x)$ ”, where  $P(x)$  is a predicate (Definition 0.9), is an element  $x$  of  $A$  for which  $P(x)$  is false. A single counterexample shows that the proposition is false.

**Example 0.11** (A counterexample). The proposition “for every integer  $n$ ,  $n + n > n$ ” is false. The integer  $n = 0$  is a counterexample:  $0 + 0 = 0$ , and  $0 > 0$  is false. The proposition is true for some integers, such as  $n = 3$ , since  $3 + 3 = 6 > 3$ , but one counterexample is enough to make it false.

**Definition 0.13** (Converse). The **converse** of the proposition “if  $P$ , then  $Q$ ” is the proposition “if  $Q$ , then  $P$ ”.

**Example 0.12** (A true proposition with a false converse). For an integer  $n$ , call  $n$  a multiple of 4 if  $n = 4k$  for some integer  $k$ .

- “If  $n$  is a multiple of 4, then  $n$  is even” is true:  $n = 4k = 2 \cdot (2k)$ , and  $2k$  is an integer.
- Its converse, “if  $n$  is even, then  $n$  is a multiple of 4”, is false. The integer  $n = 2$  is a counterexample (Definition 0.12): 2 is even, but  $2 = 4k$  only for  $k = \frac{1}{2}$ , which is not an integer.

So a proposition can be true while its converse is false.

**Definition 0.14** (Necessary and sufficient conditions). When the proposition “if  $P$ , then  $Q$ ” is true,  $P$  is a **sufficient condition** for  $Q$ , and  $Q$  is a **necessary condition** for  $P$ . When both “if  $P$ , then  $Q$ ” and its converse (Definition 0.13) are true,  $P$  is a **necessary and sufficient condition** for  $Q$ , written “ $P$  if and only if  $Q$ ”.

**Example 0.13** (Necessary but not sufficient). In Example 0.12, “if  $n$  is a multiple of 4, then  $n$  is even” is true, so:

- being a multiple of 4 is a sufficient condition for  $n$  to be even;
- being even is a necessary condition for  $n$  to be a multiple of 4.

Being even is not a sufficient condition for being a multiple of 4:  $n = 2$  is even, but not a multiple of 4.

**Definition 0.15** (Proof by contradiction). A **proof by contradiction** of a proposition  $P$  assumes that  $P$  is false, and derives from that assumption a proposition that is known to be false. Since a true assumption cannot lead to a false proposition, the assumption that  $P$  is false must itself be false, so  $P$  is true.

**Example 0.14** (No integer is both even and odd). Suppose, for contradiction, that some integer  $n$  is both even and odd (Definition 0.2). Then  $n = 2j$  and  $n = 2k + 1$  for some integers  $j$  and  $k$ , and

$$\begin{aligned} 2j &= 2k + 1 && \text{(both equal } n) \\ 2j - 2k &= 1 && \text{(subtract } 2k \text{ from both sides)} \\ 2(j - k) &= 1 && \text{(distributive law)} \\ j - k &= \frac{1}{2} && \text{(divide both sides by 2)} \end{aligned}$$

The third line uses the **distributive law**. But  $j - k$  is an integer, and  $\frac{1}{2}$  is not, so the last line is false. So no integer is both even and odd. For example,  $6 = 2 \cdot 3$  is even, and  $6 = 2k + 1$  would need  $k = 2.5$ , which is not an integer.

## Indicator functions

An **indicator function** is a mathematical function that signals whether an element belongs to a specified set, or whether a given logical condition is satisfied. In statistics and epidemiology, indicator functions are ubiquitous: they represent binary variables, censor and event indicators

in survival analysis, membership in subpopulations, and domain restrictions in integrals and sums.

Despite their conceptual simplicity, notation for indicator functions varies substantially across textbooks, research papers, and subfields. This section summarizes the principal notational conventions.

**Definition 0.16** (Indicator function). For any subset  $A \subseteq \Omega$  of a universal set  $\Omega$ , the **indicator function** of  $A$  is the function  $\mathbb{1}_A : \Omega \rightarrow \{0, 1\}$  defined by:

$$\mathbb{1}_A(x) \stackrel{\text{def}}{=} \begin{cases} 1, & x \in A \\ 0, & x \notin A \end{cases}$$

More generally, for any logical proposition or predicate  $P$ , the **indicator** of  $P$  takes the value 1 when  $P$  is true and 0 when  $P$  is false:

$$\mathbb{1}(P) \stackrel{\text{def}}{=} \begin{cases} 1, & \text{if } P \text{ is true} \\ 0, & \text{if } P \text{ is false} \end{cases}$$

**Example 0.15** (Evaluating set and predicate indicators). Consider the real line  $\Omega = \mathbb{R}$ , the set of nonnegative numbers  $A = [0, \infty)$ , and a continuous **random variable**  $Y$ .

1. **Set indicator**  $\mathbb{1}_A(x)$ :

- For  $x = 3.5$ : since  $3.5 \in [0, \infty)$ ,  $\mathbb{1}_A(3.5) = 1$ .
- For  $x = -2.1$ : since  $-2.1 \notin [0, \infty)$ ,  $\mathbb{1}_A(-2.1) = 0$ .

2. **Predicate indicator**  $\mathbb{1}(Y > 5)$ :

- If an observation yields  $Y = 7.2$ , the predicate  $7.2 > 5$  is true, so  $\mathbb{1}(7.2 > 5) = 1$ .
- If an observation yields  $Y = 4.1$ , the predicate  $4.1 > 5$  is false, so  $\mathbb{1}(4.1 > 5) = 0$ .

## Two primary notational paradigms: set vs. predicate notation

The vast majority of indicator notations belong to one of two families: **set notation** or **predicate notation**.

### Set notation

In **set notation**, the indicator is tied to a set  $A$ , which appears as a subscript:

- $\mathbf{1}_A(x)$  (bold numeral one)

- $\mathbb{1}_A(x)$  (blackboard bold numeral one)
- $I_A(x)$  (capital letter  $I$ )
- $\mathbb{I}_A(x)$  (blackboard bold letter  $I$ )
- $\chi_A(x)$  (Greek letter chi, historically termed the *characteristic function*)

When the function is viewed as a mathematical object in its own right (for instance, as an element of an  $L^p$  function space), authors often omit the argument  $x$ , writing simply  $\mathbf{1}_A$ ,  $\mathbb{1}_A$ , or  $I_A$ .

### Predicate notation

In **predicate notation**, the indicator takes a logical condition, relation, or proposition  $P$  directly as its argument or subscript:

- $\mathbb{I}(x \in A)$  or  $\mathbb{I}(P)$
- $\mathbf{1}(x \in A)$  or  $\mathbf{1}(P)$
- $\mathbb{1}(x \in A)$  or  $\mathbb{1}(P)$
- $\mathbb{1}\{x \in A\}$  or  $\mathbb{1}\{P\}$
- $I(x \in A)$  or  $I(P)$

Predicate notation is especially common in applied statistics and survival analysis, where indicators frequently depend on inequalities involving random variables, such as  $\mathbb{1}(T_i \leq t)$  (an event occurring before time  $t$ ) or  $\mathbb{1}(Y_i = 1)$  (a binary outcome).

### Equivalence between paradigms

The two paradigms are connected by evaluating the predicate indicator at the membership statement  $x \in A$ :

$$\mathbf{1}_A(x) = \mathbb{I}(x \in A)$$

Set notation is more natural when the underlying set  $A$  has a standard name (such as the support of a distribution or a geometric region). Predicate notation is more natural when the condition involves compound inequalities, such as  $\mathbb{1}(0 \leq t \leq u)$ .

### Iverson bracket notation

In 1962, Kenneth Iverson introduced a compact notation in the programming language APL, later popularized in mathematics and computer science by Donald Knuth: the [Iverson bracket](#).

**Definition 0.17** (Iverson bracket). For any logical proposition  $P$ , the **Iverson bracket** of  $P$  is

$$[P] \stackrel{\text{def}}{=} \begin{cases} 1, & \text{if } P \text{ is true} \\ 0, & \text{if } P \text{ is false} \end{cases}$$

**Example 0.16** (Evaluating Iverson brackets).

- $[3 > 2] = 1$ , because  $3 > 2$  is true.
- $[2 > 3] = 0$ , because  $2 > 3$  is false.
- $[4 \in \{1, 2\}] = 0$ , because 4 is not an element of  $\{1, 2\}$ .

*Remark 0.3* (The Iverson bracket is the predicate indicator). The Iverson bracket  $[P]$  is the predicate indicator  $\mathbb{1}(P)$  (Definition 0.16) in different notation. Under this notation, set membership is written  $[x \in A]$ , so  $[x \in A] = \mathbb{1}_A(x)$ . For example, the values in Example 0.15 become  $[7.2 > 5] = \mathbb{1}(7.2 > 5) = 1$  and, with  $A = [0, \infty)$ ,  $[-2.1 \in A] = \mathbb{1}_A(-2.1) = 0$ .

**Definition 0.18** (Kronecker delta). For integers  $i$  and  $j$ , the **Kronecker delta** is

$$\delta_{ij} \stackrel{\text{def}}{=} [i = j]$$

that is,  $\delta_{ij} = 1$  when  $i = j$  and  $\delta_{ij} = 0$  when  $i \neq j$  (Definition 0.17).

**Example 0.17** (Evaluating the Kronecker delta).

- $\delta_{11} = [1 = 1] = 1$ .
- $\delta_{12} = [1 = 2] = 0$ .
- $\delta_{22} = [2 = 2] = 1$ .
- $\delta_{23} = [2 = 3] = 0$ .

## Strengths and limitations of the Iverson bracket

The primary advantage of the Iverson bracket is algebraic conciseness: it turns a sum or integral over a subset into a sum or integral over a larger, fixed set. For example, if  $U$  is a [finite set](#) that contains  $A$ , then:

$$\sum_{x \in A} f(x) = \sum_{x \in U} f(x)[x \in A]$$

With  $A = \{2, 4\}$ ,  $f(x) = x$ , and  $U = \{1, 2, 3, 4, 5\}$ , both sides equal  $2 + 4 = 6$ .

However, in statistics and epidemiology, square brackets are already heavily overloaded: they denote [closed intervals](#)  $[a, b]$ , conditional expectations  $E[Y \mid X]$ , and matrix delimiters. To prevent visual confusion with expectation brackets or intervals, statistical literature predominantly uses  $\mathbb{1}$  or  $I$  rather than the bare Iverson bracket.

## Summary of indicator notations

Table 2 compares the major notations encountered across the literature.

Table 2: Notations for indicator functions across mathematical and statistical literature

| Notation style                        | Typical syntax                       | Primary fields                              | Notes and potential ambiguities                                                             |
|---------------------------------------|--------------------------------------|---------------------------------------------|---------------------------------------------------------------------------------------------|
| <b>Blackboard bold 1</b>              | $\mathbb{1}_A(x), \mathbb{1}(P)$     | Modern probability, mathematical statistics | Unambiguous; distinct from matrices and scalars; standard in this book.                     |
| <b>Bold numeral 1</b>                 | $\mathbf{1}_A(x), \mathbf{1}(P)$     | Probability theory, measure theory          | Can be confused with a vector of ones $\mathbf{1} = (1, \dots, 1)^\top$ .                   |
| <b>Blackboard bold I</b>              | $\mathbb{I}(x \in A), \mathbb{I}(P)$ | Econometrics, machine learning, statistics  | Clear predicate notation; avoids confusion with numerals.                                   |
| <b>Letter I</b>                       | $I_A(x), I(P)$                       | Classical statistics, epidemiology          | Can be confused with the identity matrix $I$ or Fisher information $\mathcal{I}$ .          |
| <b>Iverson bracket</b>                | $[P], [x \in A]$                     | Computer science, discrete mathematics      | Very compact, but square brackets collide with intervals and expectation brackets.          |
| <b>Greek letter <math>\chi</math></b> | $\chi_A(x)$                          | Real analysis, measure theory               | Often termed “characteristic function”; collides with the Fourier transform in probability. |

## Conventions in this book

In these notes, we standardize on blackboard bold  $\mathbb{1}$  via the macros defined in `latex-macros/macros.qmd`:

- `\indic{A}` produces  $\mathbb{1}_A$  (set subscript)
- `\indicp{P}` produces  $\mathbb{1}(P)$  (predicate in parentheses)

- `\indiccb{P}` produces  $\mathbb{1}\{P\}$  (predicate in curly braces)
- `\1{P}` produces  $1_P$  (text numeral with subscript, used in legacy formulas)

Blackboard bold  $\mathbb{1}$  is preferred because it avoids all common collisions: it is visually distinct from the scalar 1, the identity matrix  $I$ , and the information matrices  $(I, \mathcal{I})$ .

## Key algebraic properties

Indicator functions translate logical operations on events into ordinary arithmetic on real numbers:

For subsets  $A$  and  $B$  of  $\Omega$ , with **complement**  $A^c \stackrel{\text{def}}{=} \Omega \setminus A$ , and for every  $x \in \Omega$ :

- **Intersection (“and”)**:  $\mathbb{1}_{A \cap B}(x) = \mathbb{1}_A(x) \cdot \mathbb{1}_B(x)$
- **Union (“or”)**:  $\mathbb{1}_{A \cup B}(x) = \mathbb{1}_A(x) + \mathbb{1}_B(x) - \mathbb{1}_A(x) \cdot \mathbb{1}_B(x)$
- **Complement (“not”)**:  $\mathbb{1}_{A^c}(x) = 1 - \mathbb{1}_A(x)$
- **Idempotence**:  $(\mathbb{1}_A(x))^2 = \mathbb{1}_A(x)$
- **Expectation gives probability**: For any event  $A$ , the **expectation** of its indicator is the probability of the event:

$$\mathbb{E}[\mathbb{1}_A] = 0 \cdot \Pr(A^c) + 1 \cdot \Pr(A) = \Pr(A)$$

This fundamental **identity** connects probability theory directly to linear expectation. It provides the mathematical foundation for empirical proportions, survival curve estimators, and regression models for binary outcomes.

## Notational shorthands

**Definition 0.19** (Notational shorthand). A **notational shorthand** is a way of writing an expression that leaves part of the expression out, such as the set a sum runs over or the limits of an integral, and relies on the reader to supply the missing part from context. The expression with every part written out is its **full form**.

**i** These notes write the full form

In these notes, we avoid notational shorthands in permanent writing and write the full form of each expression, even where a shorthand would be shorter. A reader who supplies the missing part of a shorthand from a different context than the writer intended reads a different expression than the one the writer meant. Handwritten board work is often less

complete, but it should still write out ranges and limits where it can.

We make one exception: writing two factors side by side means multiplying them, as in  $2x$  for  $2 \cdot x$  and  $ab$  for  $a \cdot b$ . We still write the multiplication sign when the side-by-side form could be misread:

- $2 \cdot 3$ , not  $23$ , which reads as twenty-three;
- $a \cdot (b + c)$  when  $a$  could be a function, since  $a(b + c)$  could mean  $a$  evaluated at  $b + c$ .

**Definition 0.20** (Range of a variable). The **range** of a variable  $x$ , written  $\mathcal{R}(x)$ , is the set of values that  $x$  can take.

For example, if  $x$  is the number of heads in two tosses of a coin, then  $\mathcal{R}(x) = \{0, 1, 2\}$ . A **random variable**  $X$  is a **function**, and the set of values  $X$  can take,  $\mathcal{R}(X)$ , is the **image** of  $X$ . These notes use “range” only for variables: for a function, “range” can mean either the image or the codomain (**Image and range**). In statistics, “the range” of a dataset can also mean its largest value minus its smallest value;  $\mathcal{R}(x)$  is a set, not that difference.

Table 3 lists shorthands you may meet in other sources, the full form each one abbreviates, and the part each one leaves out. In the table,  $x_1, \dots, x_n$  and  $a_1, \dots, a_n$  are  $n$  numbers,  $b_{ij}$  is the entry in row  $i$  and column  $j$  of an  $m \times n$  array,  $k$  is an integer with  $1 \leq k \leq n$ ,  $f$  is a **function** from  $\mathbb{R}$  to  $\mathbb{R}$ , and, in the first row,  $\mathcal{R}(x)$  is a finite set.

Table 3: Common notational shorthands and their full forms

| Shorthand                          | Full form                                          | Part left out                                                         |
|------------------------------------|----------------------------------------------------|-----------------------------------------------------------------------|
| $\sum_x f(x)$                      | $\sum_{x \in \mathcal{R}(x)} f(x)$                 | the set the sum runs over ( <b>sum over a finite set</b> )            |
| $\sum_i x_i$                       | $\sum_{i=1}^n x_i$                                 | the lower and upper limits of the index ( <b>summation notation</b> ) |
| $\sum x_i$                         | $\sum_{i=1}^n x_i$                                 | the index and its lower and upper limits                              |
| $\prod_i x_i$                      | $\prod_{i=1}^n x_i$                                | the lower and upper limits of the index                               |
| $\prod x_i$                        | $\prod_{i=1}^n x_i$                                | the index and its lower and upper limits                              |
| $\sum_{i,j} b_{ij}$                | $\sum_{i=1}^m \sum_{j=1}^n b_{ij}$                 | the limits of both indices                                            |
| $\sum_{i \neq k} a_i$              | $\sum_{i \in \{1, \dots, n\} \setminus \{k\}} a_i$ | the set $\{1, \dots, n\}$ of indices that $k$ is removed from         |
| $\int f(x) dx$ , used for a number | $\int_{-\infty}^{\infty} f(x) dx$                  | the limits of integration                                             |
| $\int f$                           | $\int_{-\infty}^{\infty} f(x) dx$                  | the limits, the variable of integration, and the differential $dx$    |

| Shorthand                          | Full form                                                                                                | Part left out                                                                |
|------------------------------------|----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| $x_{1:n}$<br>“the function $x^2$ ” | $(x_1, x_2, \dots, x_n)$<br>“the function $f : \mathbb{R} \rightarrow \mathbb{R}$<br>with $f(x) = x^2$ ” | the explicit list of entries<br>the function’s name, domain, and<br>codomain |

*Remark 0.4* ( $\int f(x) dx$  has two readings). In calculus,  $\int f(x) dx$  with no limits is not a shorthand: it is the **indefinite integral** of  $f$ , a family of functions rather than a number. Some sources, especially in probability, also write  $\int f(x) dx$  for the number  $\int_{-\infty}^{\infty} f(x) dx$ . Writing the limits out tells the reader which of the two is meant.

**Example 0.18** (Writing out an index shorthand). Let  $n = 3$  and  $(x_1, x_2, x_3) = (2, 5, 1)$ . A source that writes  $\sum_i x_i$  means  $\sum_{i=1}^3 x_i$ :

$$\begin{aligned}
 \sum_{i=1}^3 x_i &= x_1 + x_2 + x_3 && \text{(definition of summation notation)} \\
 &= 2 + 5 + 1 && \text{(substitute the values)} \\
 &= 8 && \text{(add)}
 \end{aligned}$$

## Why is notation in probability and statistics so inconsistent and disorganized?

In grad school, we are asked to learn from increasingly disorganized materials and lectures. Not coincidentally, as the amount of organization decreases, the amount of complexity increases, the amount of difficulty increases, the number of reliable references decreases, and the amount of inconsistency in notation and content increases (both between multiple references and within single references!). In other words, as you approach the cutting-edge of most fields, you start to run into content that hasn’t been fully thought through or standardized. This lack of clarity is unfortunate and undesirable, but it is understandable and inevitable.

It’s worth noting that calculus was formalized in the **1600s**, elementary algebra was formalized around **820**, and arithmetic **even earlier**. And calculus still has **several competing notation systems**. In contrast, the field of statistics only emerged in the **late 1800s and early 1900s**, so it’s not surprising that the notation and terminology is still developing. Generalized linear models were only formalized in 1972 (Nelder and Wedderburn 1972), which is very recent in terms of the **pace of scientific development**.

## Further reading

For the logical symbols and proof conventions used in these notes:

- Velleman (2019) explains how mathematical statements are built from logical connectives and quantifiers, and how to read and write them.
- Barker-Plummer et al. (2011) covers the same symbols as a formal language, with propositional and first-order logic.

For set and function notation, see the [Sets and Functions](#) page.

## References

- Barker-Plummer, Dave, Jon Barwise, and John Etchemendy. 2011. *Language, Proof and Logic*. 2nd ed. CSLI Publications. <https://www.amazon.com/dp/1575866323>.
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- Rudin, Walter. 1976. *Principles of Mathematical Analysis*. 3rd ed. International Series in Pure and Applied Mathematics. McGraw-Hill.
- Velleman, Daniel J. 2019. *How to Prove It: A Structured Approach*. 3rd ed. Cambridge University Press. <https://doi.org/10.1017/9781108539890>.